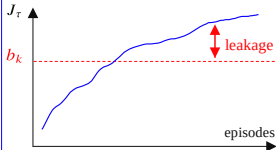


Soft penalty objective

$$\max J_\tau - \lambda^\top J_c$$

$$\text{s.t. } \tau \rightarrow \infty$$

$$J_\tau = [J_{c1}, \dots, J_{cN}]^\top, \quad \lambda = [\lambda_1, \dots, \lambda_N]^\top$$



fixed- λ penalty: some safety tasks reward b_k

weakness: constant leakage

SafeTransport

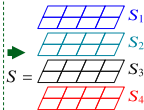
safe γ -flow bridge

1. Constraints

 Bridge S_1	0.35 1.60 0.21 0.65 1.60 0.55 0.15 1.60 0.35
 Liquid S_2	0.30 1.60 0.15 0.30 1.60 0.35 0.20 1.60 0.55
 Excavator S_3	0.20 1.60 0.12 0.25 1.60 0.25 0.20 1.60 0.35
 Fire S_4	0.15 1.60 0.13 0.20 1.60 0.15 0.20 1.60 0.15

2. Compress

uniform compression
 $S = \min_i S_i$



uniform minimum

3. Solve (OFT-Sinkhorn)

► OFT-Sinkhorn
+ PPO

$$P = \begin{bmatrix} 0.45 & 0.45 & 0.21 \\ 0.65 & 0.55 & 0.02 \\ 0.12 & 0.34 & 0.34 \end{bmatrix}$$

transport matrix

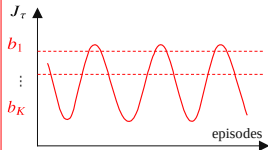
$$P^\top \leq S$$

Hard CMDP budget

$$\max J_\tau \quad \text{s.t.} \quad J_c \leq b$$

$$\tau \rightarrow 0$$

$$J_\tau = [J_{c1}, \dots, J_{cN}]^\top, \quad b = [b_1, \dots, b_N]^\top$$



dual updates: safety costs oscillate around b_k

weakness: oscillation / brittleness

$\tau = \infty$

PPO / SAC

FOCOPS

SafeTransport

CPO

hard-constraint / CUP

$\tau = 0$

soft penalty

hard budget